AP PRECALCULUS · TOPIC 1.13

Function Model Selection and Assumption Articulation

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 A table gives $y = 5, 9, 13, 17, 21$ for $x = 0, 1, 2, 3, 4$. Which model fits the data?

A. Linear, $y = 4x + 5$
B. Quadratic, $y = 4x^2 + 5$
C. Linear, $y = 5x + 4$
D. Linear, $y = 9x + 5$

2 A table gives $y = 3, 4, 9, 18, 31$ for $x = 0, 1, 2, 3, 4$. Which function fits the data?

A. $y = 4x^2 - x + 3$
B. $y = 2x^2 + x + 3$
C. $y = 2x^2 - x + 3$
D. $y = 2x^2 - x + 4$

3 A table gives $y = 2, 3, 10, 29, 66, 127$ for $x = 0, 1, 2, 3, 4, 5$. The nth differences are constant. What is n?

A. 2
B. 3
C. 5
D. 6

4 A rectangle has perimeter 60 m and width w. Which function type best models its area $A(w)$?

A. Linear
B. Cubic
C. Piecewise-defined
D. Quadratic

5 A rectangle has perimeter 60 m and width w, so its area is $A(w) = w(30 - w)$. Which domain restriction fits the context?

A. All real numbers
B. $w > 0$
C. $0 < w < 30$
D. $0 < w < 60$

6 Eight points on a graph have distinct input values. A polynomial function of degree at most n passes through all eight points. What is the least value of n that is guaranteed?

A. 8
B. 7
C. 6
D. 4

7 Which situation is best modeled by a piecewise-defined function?

A. A taxi fare of 3 dollars for the first 2 km, then 1.50 dollars for each additional km
B. The distance driven at a constant 60 km/h
C. The area of a square of side s
D. The volume of a cube of edge s

8 A model for tickets sold is $T(p) = 200 - 8p$, where p is the ticket price in dollars. Which assumption is built into the model?

A. Sales stay at 200 tickets whatever the price
B. Sales increase as the price increases
C. Prices must be whole numbers of dollars
D. Each 1 dollar increase in price lowers sales by 8 tickets

AP PRECALCULUS · TOPIC 1.13 · SECTION 2

Function Model Selection and Assumption Articulation

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

The number of visitors, in hundreds, to a museum exhibit on day x after it opens is recorded: $x = 0, 1, 2, 3, 4, 5$ gives $y = 12, 19, 24, 27, 28, 27$.

(a) Show that a quadratic model is appropriate for these data. DOK3 2 pt

(b) Find a quadratic model and use it to find the greatest number of visitors and the day it occurs. DOK3 2 pt

(c) The model is used until $y = 0$. Find that value of x to three decimal places and state one reason the model should not be used beyond it. DOK3 2 pt

Question 10 No calculator · 6 points

An open box is made from an 18 cm by 18 cm square sheet by cutting a square of side x cm from each corner and folding up the sides, so $V(x) = x(18 - 2x)^2$.

(a) State a domain for V that fits the context and give the reason for each end. DOK3 2 pt

(b) State the type and degree of V and justify the choice using the context. DOK3 2 pt

(c) A different data set has $(0, 1),\ (1, 2),\ (2, 5),\ (3, 10)$. Use differences to find the type, then find y when $x = 5$. DOK3 2 pt